

B.4 Graphs

This section presents two kinds of graphs: directed and undirected. Certain definitions in the literature differ from those given here, but for the most part, the differences are slight. Section 20.1 shows how to represent graphs in computer memory.

A **directed graph** (or **digraph**) G is a pair (V, E) , where V is a finite set and E is a binary relation on V . The set V is called the **vertex set** of G , and its elements are called **vertices** (singular: **vertex**). The set E is called the **edge set** of G , and its elements are called **edges**. Figure B.2(a) is a pictorial representation of a directed graph on the vertex set $\{1, 2, 3, 4, 5, 6\}$. Vertices are represented by circles in the figure, and edges are represented by arrows. **Self-loops**—edges from a vertex to itself—are possible.

In an **undirected graph** $G = (V, E)$, the edge set E consists of *unordered* pairs of vertices, rather than ordered pairs. That is, an edge is a set $\{u, v\}$, where $u, v \in V$ and $u \neq v$. By convention, we use the notation (u, v) for an edge, rather than the set notation $\{u, v\}$, and we consider (u, v) and (v, u) to be the same edge. In an undirected graph, self-loops are forbidden, so that every edge consists of two distinct vertices. Figure B.2(b) shows an undirected graph on the vertex set $\{1, 2, 3, 4, 5, 6\}$.

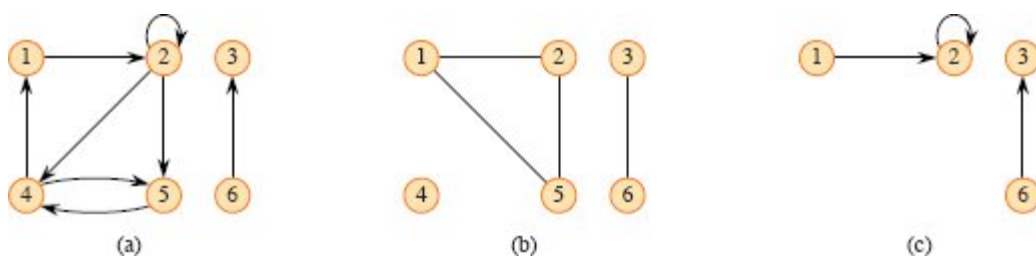


Figure B.2 Directed and undirected graphs. **(a)** A directed graph $G = (V, E)$, where $V = \{1, 2, 3, 4, 5, 6\}$ and $E = \{(1, 2), (2, 2), (2, 4), (2, 5), (4, 1), (4, 5), (5, 4), (6, 3)\}$. The edge $(2, 2)$ is a self-loop. **(b)** An undirected graph $G = (V, E)$, where $V = \{1, 2, 3, 4, 5, 6\}$ and $E = \{(1, 2), (1, 5), (2, 5), (3, 6)\}$. The vertex 4 is isolated. **(c)** The subgraph of the graph in part (a) induced by the vertex set $\{1, 2, 3, 6\}$.

Many definitions for directed and undirected graphs are the same, although certain terms have slightly different meanings in the two

contexts. If (u, v) is an edge in a directed graph $G = (V, E)$, we say that (u, v) is *incident from* or *leaves* vertex u and is *incident to* or *enters* vertex v . For example, the edges leaving vertex 2 in Figure B.2(a) are $(2, 2)$, $(2, 4)$, and $(2, 5)$. The edges entering vertex 2 are $(1, 2)$ and $(2, 2)$. If (u, v) is an edge in an undirected graph $G = (V, E)$, we say that (u, v) is *incident on* vertices u and v . In Figure B.2(b), the edges incident on vertex 2 are $(1, 2)$ and $(2, 5)$.

If (u, v) is an edge in a graph $G = (V, E)$, we say that vertex v is *adjacent* to vertex u . When the graph is undirected, the adjacency relation is symmetric. When the graph is directed, the adjacency relation is not necessarily symmetric. If v is adjacent to u in a directed graph, we can write $u \rightarrow v$. In parts (a) and (b) of Figure B.2, vertex 2 is adjacent to vertex 1, since the edge $(1, 2)$ belongs to both graphs. Vertex 1 is *not* adjacent to vertex 2 in Figure B.2(a), since the edge $(2, 1)$ is absent.

The *degree* of a vertex in an undirected graph is the number of edges incident on it. For example, vertex 2 in Figure B.2(b) has degree 2. A vertex whose degree is 0, such as vertex 4 in Figure B.2(b), is *isolated*. In a directed graph, the *out-degree* of a vertex is the number of edges leaving it, and the *in-degree* of a vertex is the number of edges entering it. The *degree* of a vertex in a directed graph is its in-degree plus its out-degree. Vertex 2 in Figure B.2(a) has in-degree 2, out-degree 3, and degree 5.

A *path* of *length* k from a vertex u to a vertex u' in a graph $G = (V, E)$ is a sequence $\langle v_0, v_1, v_2, \dots, v_k \rangle$ of vertices such that $u = v_0$, $u' = v_k$, and $(v_{i-1}, v_i) \in E$ for $i = 1, 2, \dots, k$. The length of the path is the number of edges in the path, which is 1 less than the number of vertices in the path. The path *contains* the vertices $v_0, v_1, \dots, v_k$ and the edges $(v_0, v_1), (v_1, v_2), \dots, (v_{k-1}, v_k)$. (There is always a 0-length path from u to u .) If there is a path p from u to u' , we say that u' is *reachable* from u via p , which we can write as $u \rightsquigarrow u'$. A path is *simple*⁴ if all vertices in the path are distinct. In Figure B.2(a), the path $\langle 1, 2, 5, 4 \rangle$ is a simple path of length 3. The path $\langle 2, 5, 4, 5 \rangle$ is not simple. A *subpath* of path $p = \langle v_0,$

$v_1, \dots, v_k$ is a contiguous subsequence of its vertices. That is, for any $0 \leq i \leq j \leq k$, the subsequence of vertices $\langle v_i, v_{i+1}, \dots, v_j \rangle$ is a subpath of p .

In a directed graph, a path $\langle v_0, v_1, \dots, v_k \rangle$ forms a **cycle** if $v_0 = v_k$ and the path contains at least one edge. The cycle is **simple** if, in addition, $v_1, v_2, \dots, v_k$ are distinct. A cycle consisting of k vertices has **length** k . A self-loop is a cycle of length 1. Two paths $\langle v_0, v_1, v_2, \dots, v_{k-1}, v_0 \rangle$ and $\langle v'_0, v'_1, v'_2, \dots, v'_{k-1}, v'_0 \rangle$ form the same cycle if there exists an integer j such that $v'_i = v_{(i+j) \bmod k}$ for $i = 0, 1, \dots, k-1$. In Figure B.2(a), the path $\langle 1, 2, 4, 1 \rangle$ forms the same cycle as the paths $\langle 2, 4, 1, 2 \rangle$ and $\langle 4, 1, 2, 4 \rangle$. This cycle is simple, but the cycle $\langle 1, 2, 4, 5, 4, 1 \rangle$ is not. The cycle $\langle 2, 2 \rangle$ formed by the edge $(2, 2)$ is a self-loop. A directed graph with no self-loops is **simple**. In an undirected graph, a path $\langle v_0, v_1, \dots, v_k \rangle$ forms a **cycle** if $k > 0$, $v_0 = v_k$, and all edges on the path are distinct. The cycle is **simple** if $v_1, v_2, \dots, v_k$ are distinct. For example, in Figure B.2(b), the path $\langle 1, 2, 5, 1 \rangle$ is a simple cycle. A graph with no simple cycles is **acyclic**.

An undirected graph is **connected** if every vertex is reachable from all other vertices. The **connected components** of an undirected graph are the equivalence classes of vertices under the “is reachable from” relation. The graph shown in Figure B.2(b) has three connected components: $\{1, 2, 5\}$, $\{3, 6\}$, and $\{4\}$. Every vertex in the connected component $\{1, 2, 5\}$ is reachable from every other vertex in $\{1, 2, 5\}$. An undirected graph is connected if it has exactly one connected component. The edges of a connected component are those that are incident on only the vertices of the component. In other words, edge (u, v) is an edge of a connected component only if both u and v are vertices of the component.

A directed graph is **strongly connected** if every two vertices are reachable from each other. The **strongly connected components** of a directed graph are the equivalence classes of vertices under the “are mutually reachable” relation. A directed graph is strongly connected if it has only one strongly connected component. The graph in Figure B.2(a) has three strongly connected components: $\{1, 2, 4, 5\}$, $\{3\}$, and

$\{6\}$. All pairs of vertices in $\{1, 2, 4, 5\}$ are mutually reachable. The vertices $\{3, 6\}$ do not form a strongly connected component, since vertex 6 cannot be reached from vertex 3.

Two graphs $G = (V, E)$ and $G' = (V', E')$ are *isomorphic* if there exists a bijection $f: V \rightarrow V'$ such that $(u, v) \in E$ if and only if $(f(u), f(v)) \in E'$. In other words, G and G' are isomorphic if the vertices of G can be relabeled to be vertices of G' , maintaining the corresponding edges in G and G' . Figure B.3(a) shows a pair of isomorphic graphs G and G' with respective vertex sets $V = \{1, 2, 3, 4, 5, 6\}$ and $V' = \{u, v, w, x, y, z\}$. The mapping from V to V' given by $f(1) = u, f(2) = v, f(3) = w, f(4) = x, f(5) = y, f(6) = z$ provides the required bijective function. The graphs in Figure B.3(b) are not isomorphic. Although both graphs have 5 vertices and 7 edges, the top graph has a vertex of degree 4 and the bottom graph does not.

We say that a graph $G' = (V', E')$ is a *subgraph* of $G = (V, E)$ if $V' \subseteq V$ and $E' \subseteq E$. Given a set $V' \subseteq V$, the subgraph of G *induced* by V' is the graph $G' = (V', E')$, where

$$E' = \{(u, v) \in E : u, v \in V'\}.$$

The subgraph induced by the vertex set $\{1, 2, 3, 6\}$ in Figure B.2(a) appears in Figure B.2(c) and has the edge set $\{(1, 2), (2, 2), (6, 3)\}$.

Given an undirected graph $G = (V, E)$, the *directed version* of G is the directed graph $G' = (V, E')$, where $(u, v) \in E'$ if and only if $(u, v) \in E$. That is, each undirected edge (u, v) in G turns into two directed edges, (u, v) and (v, u) , in the directed version. Given a directed graph $G = (V, E)$, the *undirected version* of G is the undirected graph $G' = (V, E')$, where $(u, v) \in E'$ if and only if $u \neq v$ and E contains at least one of the edges (u, v) and (v, u) . That is, the undirected version contains the edges of G “with their directions removed” and with self-loops eliminated. (Since (u, v) and (v, u) are the same edge in an undirected graph, the undirected version of a directed graph contains it only once, even if the directed graph contains both edges (u, v) and (v, u) .) In a directed graph $G = (V, E)$, a *neighbor* of a vertex u is any vertex that is adjacent to u in the undirected version of G . That is, v is a neighbor of u if $u \neq v$

and either $(u, v) \in E$ or $(v, u) \in E$. In an undirected graph, u and v are neighbors if they are adjacent.

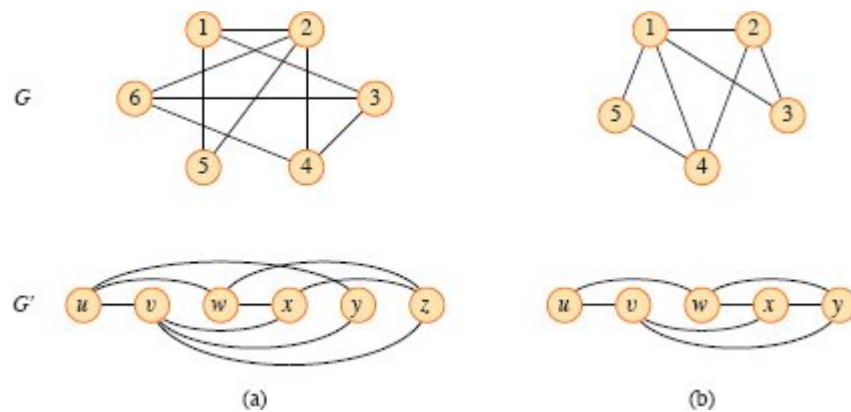


Figure B.3 (a) A pair of isomorphic graphs. The vertices of the top graph are mapped to the vertices of the bottom graph by $f(1) = u, f(2) = v, f(3) = w, f(4) = x, f(5) = y, f(6) = z$. **(b)** Two graphs that are not isomorphic. The top graph has a vertex of degree 4, and the bottom graph does not.

Several kinds of graphs have special names. A **complete graph** is an undirected graph in which every pair of vertices is adjacent. An undirected graph $G = (V, E)$ is **bipartite** if V can be partitioned into two sets V_1 and V_2 such that $(u, v) \in E$ implies either $u \in V_1$ and $v \in V_2$ or $u \in V_2$ and $v \in V_1$. That is, all edges go between the two sets V_1 and V_2 . An acyclic, undirected graph is a **forest**, and a connected, acyclic, undirected graph is a **(free) tree** (see Section B.5). We often take the first letters of “directed acyclic graph” and call such a graph a **dag**.

There are two variants of graphs that you may occasionally encounter. A **multigraph** is like an undirected graph, but it can have both multiple edges between vertices (such as two distinct edges (u, v) and (u, v)) and self-loops. A **hypergraph** is like an undirected graph, but each **hyperedge**, rather than connecting two vertices, connects an arbitrary subset of vertices. Many algorithms written for ordinary directed and undirected graphs can be adapted to run on these graphlike structures.

The **contraction** of an undirected graph $G = (V, E)$ by an edge $e = (u, v)$ is a graph $G' = (V', E')$, where $V' = V - \{u, v\} \cup \{x\}$ and x is a new vertex. The set of edges E' is formed from E by deleting the edge (u, v)

and, for each vertex w adjacent to u or v , deleting whichever of (u, w) and (v, w) belongs to E and adding the new edge (x, w) . In effect, u and v are “contracted” into a single vertex.

Exercises

B.4-1

Attendees of a faculty party shake hands to greet each other, with every pair of professors shaking hands one time. Each professor remembers the number of times he or she shook hands. At the end of the party, the department head asks the professors for their totals and adds them all up. Show that the result is even by proving the *handshaking lemma*: if $G = (V, E)$ is an undirected graph, then

$$\sum_{v \in V} \text{degree}(v) = 2|E|.$$

B.4-2

Show that if a directed or undirected graph contains a path between two vertices u and v , then it contains a simple path between u and v . Show that if a directed graph contains a cycle, then it contains a simple cycle.

B.4-3

Show that any connected, undirected graph $G = (V, E)$ satisfies $|E| \geq |V| - 1$.

B.4-4

Verify that in an undirected graph, the “is reachable from” relation is an equivalence relation on the vertices of the graph. Which of the three properties of an equivalence relation hold in general for the “is reachable from” relation on the vertices of a directed graph?

B.4-5

What is the undirected version of the directed graph in Figure B.2(a)? What is the directed version of the undirected graph in Figure B.2(b)?

B.4-6

Show how a bipartite graph can represent a hypergraph by letting incidence in the hypergraph correspond to adjacency in the bipartite graph. (*Hint:* Let one set of vertices in the bipartite graph correspond to vertices of the hypergraph, and let the other set of vertices of the bipartite graph correspond to hyperedges.)

B.5 Trees

As with graphs, there are many related, but slightly different, notions of trees. This section presents definitions and mathematical properties of several kinds of trees. Sections 10.3 and 20.1 describe how to represent trees in computer memory.

B.5.1 Free trees

As defined in Section B.4, a *free tree* is a connected, acyclic, undirected graph. We often omit the adjective “free” when we say that a graph is a tree. If an undirected graph is acyclic but possibly disconnected, it is a *forest*. Many algorithms that work for trees also work for forests. Figure B.4(a) shows a free tree, and Figure B.4(b) shows a forest. The forest in Figure B.4(b) is not a tree because it is not connected. The graph in Figure B.4(c) is connected but neither a tree nor a forest, because it contains a cycle.

The following theorem captures many important facts about free trees.

Theorem B.2 (Properties of free trees)

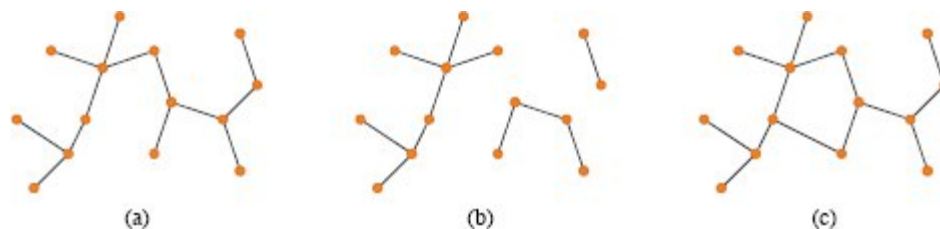


Figure B.4 (a) A free tree. (b) A forest. (c) A graph that contains a cycle and is therefore neither a tree nor a forest.